

INSPECTION AND TEST PLAN

Section 1600

SHAFT SEAL ASSEMBLY

Project: SEVEN MILE

Unit 4

GE Hydro		INSPECTION AND TEST PLAN			
Project Seven Mile		Customer: B.C. Hydro		Contract: SM-70	
Unit 4		Section 1600: Shaft seal assembly			
Page: 1601					

CODE			
A: Hold point/ customer	I : Inspection and test point R: Results recorded in Q-A files Copy to customer	M: Monitoring in process O: Assembly Operation V: In process Verification/ operation	
T: Witness point/ customer			

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
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1601	100	I,R	Incoming inspection.	See ITP Section 100		QCR 101	
1602	2100	O,M,R	Clean the head cover seal face, O'ring groove and threaded holes. Install the O'ring in the groove. Place the shaft seal base ring segments around the shaft. Assemble the base ring segments. Torque. Torque monitoring.	Installation proc. SMI-U04-02057 steps 2.1 to 2.3 Fugesco's manual C-0276-01 section 4.3 Drawing 222038220	Torque value is 355 N.m lub.	QCR 004	 Jan 17 th 2003

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: November 25 th , 2002	Accepted by: Date:	Revision no: 00
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 GE Hydro		INSPECTION AND TEST PLAN			
Project Seven Mile		Customer: B.C. Hydro		Contract: SM-70	
Unit 4		Section 1600: Shaft seal assembly			
Page: 1602					

CODE			
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1603	2100	O	Orient the assembled ring with reference to the intake centreline and centre to the shaft.	Installation proc. SMI-U04-02057 step 2.4 Fugesco's manual C-0276-01 section 4.3 Drawing 222038220			 Jan 15th 2003
1604	2100	O	Dimensional inspection: -Concentricity of shaft seal in relation to turbine shaft	Installation proc. SMI-U04-02057 step 2.4 Fugesco's manual C-0276-01		QCR 2101	 Jan 15th 2003

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GE Hydro

INSPECTION AND TEST PLAN

Project Seven Mile

Customer: B.C. Hydro

Contract: SM-70

Unit 4

Section 1600: Shaft seal assembly

Page: 1603

CODE

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Copy to customer

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Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
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1605	2100	O,M,R	Install the base ring studs and nuts. Torque. Torque monitoring.	section 4.3 Drawing 222038220 Installation proc. SMI-U04-02057 step 2.5 Fugesco's manual C-0276-01 section 4.3 Drawings 222038220 222038211	Torque value is 1520 450 N.m lub.	QCR 004	 Jan 30 th 2003
1606	2100	O	Drill and ream from the base ring into the head cover four holes dowel pins. Install the dowel pins.	Installation proc. SMI-U04-02057 step 2.6 Fugesco's manual			 Feb 12 th 2003

Prepared by:

Bruno Auger, Eng.

Approved by: Bruno Auger, Eng.

Date: November 25th, 2002

Accepted by:

Date:

Revision

no: 00

 GE Hydro		INSPECTION AND TEST PLAN			
Project Seven Mile		Customer: B.C. Hydro		Contract: SM-70	
Unit 4		Section 1600: Shaft seal assembly			
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		Page: 1604			

Oper. #	Ref. GE	Code	Description	Reference documents	Comments	Form	Date and initials
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				C-0276-01 section 4.3 Drawing 222038220			
1607	2100	O,I,R	Continue mounting the seal parts as per Fugesco's manual. Perform inspection as required in the manual.	Installation proc. SMI-U04-02057 step 2.7 Fugesco's manual C-0276-01 sections 4.4 to 4.15 Drawing 222038220		QCR 004 /	 Feb 21st 2003
1608	2100	I,R	Measure the wear indicator length.	Installation proc. SMI-U04-02057 step 2.8		QCR 004	 03/12/03

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: November 25 th , 2002	Accepted by: Date:	Revision no: 00
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GE Hydro		INSPECTION AND TEST PLAN			
Project Seven Mile		Customer: B.C. Hydro		Contract: SM-70	
Unit 4		Section 1600: Shaft seal assembly		Page: 1605	
CODE A: Hold point/ customer T: Witness point/ customer		I : Inspection and test point R: Results recorded in Q-A files Copy to customer		M: Monitoring in process O: Assembly Operation V: In process Verification/ operation	
Oper. #	Ref. GE	Code	Description	Reference documents	Comments
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1609	2100	I,R	Final inspection. Ensure that all work and documentation have been completed.	Fugesco's manual C-0276-01 section 5.1 Drawing 222038220 Installation proc. SMI-U04-02057	QCR 003 03/e4/03

RECEIVED
 03/e4/03
 CH

Prepared by: Bruno Auger, Eng.	Approved by: Bruno Auger, Eng. Date: November 25 th , 2002	Accepted by: Date:	Revision no: 00
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PROJECT: Seven Mile

Unit: 4

ITP step no.: 1602

EQUIPMENT : Torque on shaft seal base ring segments

The torque used to hold the shaft seal base ring segments together is 355 N. m lubricated.



Measured by.: Jake Zander van Jan 16th 2023

GE Repres.: [Signature] 1/17/2003

Cust. Repres.: [Signature] (/ /)

GE Canada
International Inc.

Reference NC no.: _____

Comments: _____

Rev.: 0

Date: 07/23/01



CONCENTRICITY OF SHAFT SEAL IN RELATION TO TURBINE SHAFT

QCR 2101

PROJECT: Seven Mile

Unit: 4

ITP step no.: 1604

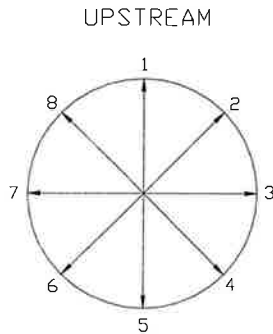
Measurement Instruments

Feeler gauge

Ambient Temperature

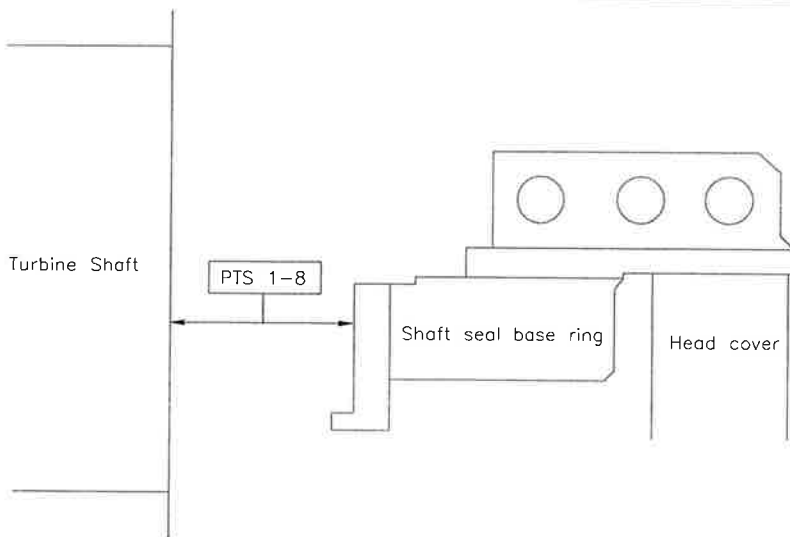
Before: _____

After: _____



PTS	GAP (mm)
1	3.30
2	3.15
3	2.92
4	2.92
5	3.05
6	3.18
7	3.23
8	3.40

	TOLERANCE (mm)	CALCULATED (mm)
CONCENTRICITY	0.8	0.22



Measured by.: Rob Speers (Jan 13th 2003)
GE Repres.: Don Burrill
Cust. Repres.: Julio Santana (Jan 15th 2003)

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JAN 15 2003
OK

GE Canada
International Inc.

Reference NC no.: _____
Comments: _____

Rev.: 0
Date: 02-11-25



GENERAL INSPECTION REPORT

QCR 004

PROJECT: Seven Mile

Unit: 4

ITP step no.: 1605

EQUIPMENT : Torquing of shaft seal base ring to headcover

The nuts holding the shaft seal base ring to the headcover have been torqued at a value of 450 N.m lubricated.

Measured by.: P. Montoya (/ /)GE Repres.: Philippe Montoya (Jan 29th 2003)

Cust. Repres.: _____ (/ /)

GE Canada
International Inc.

Reference NC no.: _____

Comments: _____

Rev.: 0

Date: 07/23/01



GENERAL INSPECTION REPORT

QCR 004

PROJECT: Seven Mile

Unit: 4

ITP step no.: 1607

EQUIPMENT : Shaft seal installation

All the steps of the Fugesco manual were followed during the shaft seal installation. All the torques used were those of the manuel.

RECEIVED
FEB 21 2003

CH

Measured by.: D. Lavontagne (/ /)
GE Repres.: Jake Zondervan
Cust. Repres.: D. Lavontagne (Feb 14 2003) (/ /)

GE Canada
International Inc.Reference NC no.: _____
Comments: _____Rev.: 0
Date: 07/23/01



GENERAL INSPECTION REPORT

QCR 004

PROJECT: Seven Mile

Unit: 4

ITP step no.: 1608

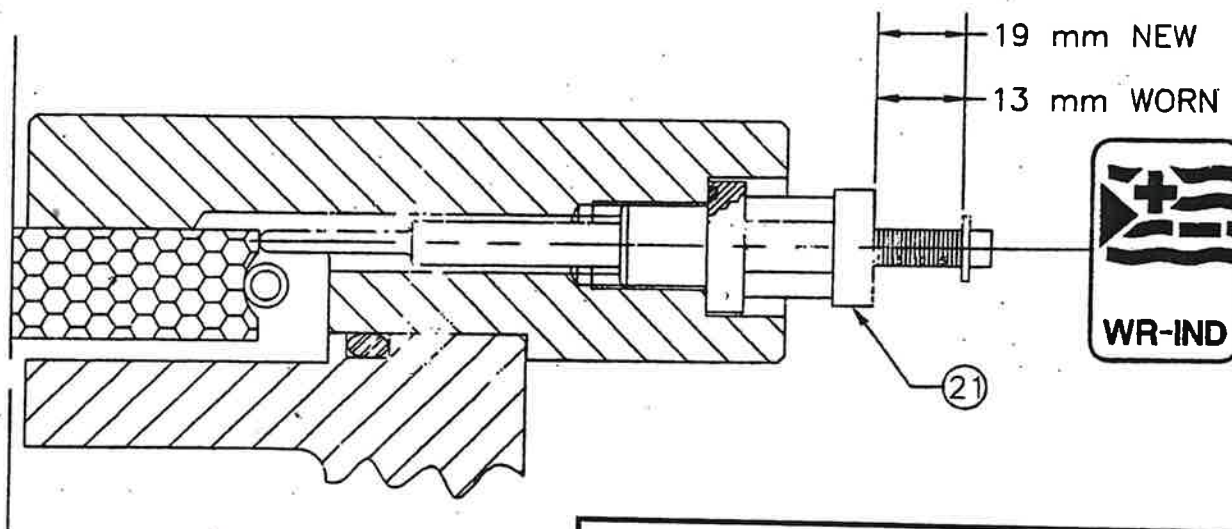
EQUIPMENT : Shaft seal wear indicator length

The shaft seal wear indicator rod sticking out has been measured.

The length is

17 mm

The rod has been measured after
Some Mechanical runs, March 5th 2003 *FB*



RECEIVED
March
11/03

Measured by.: Mark Hitchin (03/05/03)GE Repres.: Frank Bant (03/05/03)Cust. Repres.: (/ /)GE Canada
International Inc.Reference NC no.: Comments:

Rev.: 0

Date: 07/23/01



FINAL INSPECTION

QCR 003

PROJECT: Seven Mile

Unit: 4

ITP step no.: 1609

THIS FORM MUST BE COMPLETED BY THE TECHNICAL DIRECTOR, QUALITY ASSURANCE OR HIS REPRESENTATIVE.

THIS FORM MUST BE COMPLETED AT THE END OF WORK FOR EACH SECTION IN THE INSPECTION AND THE TEST PLAN.

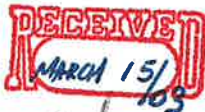
ALL NON CONFORMANCES, DISCREPANCIES, ETC. MUST BE CORRECTED BEFORE COMPLETING THIS FORM.

ONLY THE SPECIFIED ITEMS IN CONTRACT ARE SUBMITTED TO THE FINAL INSPECTION.

	YES	NO	N/A
ALL WORK WAS COMPLETED IN CONFORMITY WITH INSTALLATION MANUAL AND DRAWINGS.	✓		
THE FINAL PRODUCT HAVE BEEN INSPECTED AT THE END OF WORK AND IS IN CONFORMITY WITH CONTRACT REQUIREMENTS.	✓		
ALL NON CONFORMANCES HAVE BEEN CORRECTED ACCORDING TO THE DISPOSITION INSTRUCTIONS.	✓		
ALL CORRECTIVE ACTION REQUESTS HAVE BEEN CORRECTED AT THE CUSTOMER SATISFACTION.	✓		
ALL QUALITY ASSURANCE REPORTS HAVE BEEN COMPLETED ACCORDING TO I.T.P.	✓		
ALL FORMS HAVE BEEN REVISED AND APPROVED BY THE QUALITY ASSURANCE REPRESENTATIVE.	✓		
APPROVAL OF ENGINEERING HAVE BEEN OBTAINED WHEN NECESSARY.	✓		
APPROVAL OF CUSTOMER OR HIS REPRESENTATIVE HAVE BEEN OBTAINED WHEN NECESSARY.	✓		

REF. N.C. AND C.A.R. NB.: _____

COMMENTS : _____

_____Measured by.: Frank Benlt (03/12/03)GE Repres.: Frank Benlt (03/12/03)Cust. Repres.: [Signature] 04 Mar 2003GE Canada
International Inc.Reference NC no.: _____
Comments: _____Rev.: 0
Date: 07/23/01